

3D Human Pose Tracking using Multi-View 2D Detections

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Abstract

1 Introduction

2 Dataset Description

3 Background

This project builds directly upon concepts covered in the Computer Vision course (Lanz Spring 2026), particularly the chapters on Recognition and Object Tracking.

3.1 Object Detection

3.2 Object Tracking

3.3 Camera Calibration

3.4 Triangulation

4 Methodology

This section of the report will detail the architectures of the tracking system developed in the project. Further it covers the selected evaluation metrics and how evaluation was performed.

4.1 System Architecture

4.1.1 Overview

4.1.2 2D Detections

4.1.3 Kalman Filter

4.1.4 Track Creation and Deletion

4.1.5 Association of Detections to Tracks

4.1.6 Association Between Views

4.1.7 Auxiliary Time and Location Embeddings

4.2 Evaluation Procedure

4.3 Evaluation Metrics

5 Results and Discussion

This section of the report will present the evaluation results obtained by the developed system on the chosen dataset. It further contains also discussion of the presented results and possible reasons for the observed deficiencies.

5.1 Quantitative Results

5.2 Qualitative Results

6 Conclusion and Outlook

References

Lanz, Oswald (Spring 2026). *Computer Vision: Lecture Notes*. Course lecture notes for Free University of Bolzano course number 73075. Accessed June 10, 2026.